FLAGSHIP SOLUTION SCALING TRACK PROFILE

Scaling Climate-Smart Water and Soil Solution Bundles through Participatory and Inclusive Innovations in Guatemala and Colombia

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 1 · Close the Climate Adaptation Gap through Scalable Climate Services and Smart Water Management

Flagship 1.2 · Scaling Smart Irrigation & Climate-Resilient Water Systems

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Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management
  - **Flagship 1.1** — Digital Climate Services for Resilient Farm & Enterprise Management
  - **Flagship 1.2** — Scaling Smart Irrigation & Climate-Resilient Water Systems
- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems
  - **Flagship 2.1** — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems
  - **Flagship 2.2** — Entrepreneurship through Mechanization & Postharvest Services
  - **Flagship 2.3** — Genetics, Inputs & Services for Animals & Aquaculture
- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition
  - **Flagship 3.1** — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

Farmers in Guatemala's Dry Corridor and Colombia face a critical adaptation gap. Increasingly severe climate risks (including prolonged droughts, erratic rainfall, and soil degradation)threaten the livelihoods of over 90% of farmers who depend on rain-fed agriculture. Although climate-smart soil and water practices exist, adoption remains fragmented and limited, constrained by weak coordination among innovation system actors, finance gaps, and insufficient local capacity to design and implement scaling strategies.

This Scaling Track aims to overcome these barriers by scaling climate-smart water and soil solution bundles (CS-WS) through participatory, inclusive innovations across Guatemala and Colombia. The solution is not a single technology but a **bundle of context-specific practices** (e.g., terraces, water harvesting, drought-tolerant seeds, organic fertilizers) combined with a **participatory delivery model** that leverages existing platforms: Local Technical Agroclimatic Committees (LTACs), Climate-Smart Villages (CSVs), and the PICSA methodology. The innovation's novelty lies in its integration of validated practices that reduce climate vulnerability with delivery through champion farmers, community organizations, and financial institutions, ensuring scalability across diverse agroecological and social contexts.

The track builds on over a decade of evidence from CCAFS, AgriLAC CGIAR Regional Initiative and bilateral projects. In Guatemala, farmers adopting these bundles have reported yield gains up to 40% in maize and beans, with 70% perceiving reduced climate vulnerability. In Colombia, 70–85% of farmers reported higher income and food security. With a Scaling Readiness level of 9 and Use level of 8, the innovation is mature and field-proven. The track is implemented by the Alliance of Bioversity International and CIAT, in collaboration with IWMI and a network of government agencies, local organizations, microfinance institutions, and regional bodies.

The scaling of the CS-WS bundles is financed through the integration with ongoing large-scale investment projects, including the GCF-CSICAP project in Colombia (scaling pathway 1) and the CRS-led ASA initiative in Central America (scaling pathway 2), which allocate resources for implementation and farmer outreach but they lack of tailored responsible scaling strategies to deliver sustained impact. In addition, partnerships with microfinance institutions and cooperatives will also enable farmer-led investment and autonomous uptake of CS-WS bundles (scaling pathway 3).

Delivery at scale is executed by existing partner networks embedded within each pathway, including project implementers, extension services, local organizations, and producer associations. These actors operationalize context-specific scaling routes and deliver bundles directly to farmers through established field-based advisory systems.

S4I acts as a systemic catalyst by strengthening the effectiveness of these delivery systems. It does so by packaging validated practices into scalable bundles, co-designing pathway-specific scaling strategies, generating decision-relevant evidence, and brokering coordination among key actors to overcome fragmentation and other bottlenecks that currently limit scale. Rather than creating new delivery structures, these pathways unlocks the scaling potential of existing systems.

2. Innovation Bundle and Scaling Ambition

2.1 What is Being Scaled

The innovation is a set of climate-smart soil and water practice bundles, composed of combinations of validated agronomic and natural resource management practices. These bundles are not fixed packages; rather, they are context-specific configurations tailored to the agroecological conditions, livelihood strategies, and investment capacities of farmers, as well as to the requirements and delivery mechanisms of each scaling pathway.

The bundles draw from a common portfolio of proven practices that help to reduce climate vulnerability, including:

- **Soil conservation practices:** Terraces, living barriers, dead barriers, organic fertilizers.
- **Water management interventions:** Water reservoirs, irrigation systems (drip, micro-sprinkler), water harvesting (ditches, acequias).
- **Climate-adapted germplasm:** Drought-tolerant maize and bean seeds (e.g., ICTA Patriarca in Guatemala).
- **Integrated farming systems:** Intercropping (maize-beans), climate-smart home gardens, shade-grown coffee systems.

This modular approach allows the innovation to function as a flexible yet structured scaling unit, enabling its integration into diverse delivery systems (projects, territorial platforms, and policy instruments) while maintaining coherence in its core components and objectives.

2.2 Evidence of Impact

- **Yield increases:** Maize and bean production grew by over 300% with trenches and dead barriers in Guatemala; drought-tolerant bean varieties exceeded yields 250% above the national average (Martínez et al., 2024).
- **Soil health:** Soil organic matter increased by 0.52% to 1.98%; soil moisture improved by more than 10% on average during dry months through terraces and living barriers.
- **Economic returns:** Terraces in maize achieved IRR of 26.6% over five years; irrigation in annual crops reached IRR of 2053% in Guatemala (Add Value project, 2020).
- **Farmer perceptions:** 70–90% of women and 65–85% of men reported reduced climate vulnerability; 55–75% of women and 60–80% of men reported increased income.
- **Adoption rates:** Over 60% of farmers in the Guatemalan CSV adopted at least one practice, with 100% of women identifying yield increases from water reservoirs.

2.3 Scaling Targets

Outcome Metric	2026 Target	Long-term Goal (3–5 years)
Farmers reached (cumulative)	12,000 producers	41,000 producers (Guatemala and Colombia)
Women and youth participation	≥40% (4,800)	≥50% (20,500)
Marginalized groups reached	≥5% (600)	≥10% (4,100)
New strategic partners engaged	At least 10	15+ organizations in innovation networks
Policy influence	—	At least one national/regional policy informed by or integrating bundles
Perceived reduction in crop losses	—	15–20% reduction reported by farmers

3. Scaling Pathway and Theory of Change

3.1 Pathway Strategy

The scaling strategy is built on a **public-private-community partnership (PPP) model** that leverages three interconnected pathways, reflecting that partners are at different stages of scaling readiness:

- **Project-Based Scaling Pathway:** This pathway embeds scaling support into two large, ongoing initiatives: the **GCF-financed CSICAP project in Colombia** (focusing on CSA across multiple crops) and the **CRS-led ASA project** (Agua y Suelo para la Agricultura in spanish) across Central America's Dry Corridor. These projects have guaranteed funding and a mandate to scale but their scaling strategy is nonexistent or limitedS4I provides the science-based scaffolding (packaging bundles using IPSR, co-designing scaling strategies, and strengthening MEL) to significantly enhance their probability of success. This pathway represents the primary scaling engine, accounting for the majority of farmer reach.
- **Multi-Stakeholder Driven Scaling (LTAC Network):** This pathway leverages the established network of **77 Local Technical Agroclimatic Committees (LTACs) across 11 countries** (initially focused on Colombia and Guatemala.). These institutionalized, territorially grounded platforms already generate monthly agroclimatic recommendations. S4I will work with LTACs to systematically "harvest" farmer-prioritized practices, package them into scalable bundles, and use the LTAC network itself as a delivery mechanism, connecting climate information directly to on-farm practice adoption. This pathway plays a critical role in contextual adaptation and decentralized scaling through territorial networks.
- **Policy and Institutional Pathway:** This pathway focuses on integrating validated bundles into national and regional policy frameworks. In Colombia, this aligns with the NDC target of reaching 1 million farmers with CSA practices. In Guatemala, it supports the National Irrigation Policy (2024-2033) and the NAP-Ag. S4I will provide the evidence base, technical inputs, and policy engagement to help ministries (MAGA, MADR, MARN, MADS) and regional bodies (CAC) incorporate these bundles into public programs, extension systems, and financing mechanisms, ensuring institutional ownership beyond the project's lifespan. This pathway ensures long-term institutionalization, though with slower and more indirect scaling effects.

These pathways are not siloed; lessons from project-based scaling inform LTAC recommendations, and evidence from both feeds into policy dialogue, creating a reinforcing ecosystem for sustainable scaling.

Across all pathways, scaling is achieved by embedding solution bundles into existing delivery systems that already operate at scale. Implementation is financed by partner-led investments and delivered through their established technical and extension capacities, while S4I provides the strategic, analytical, and coordination support required to transform these channels into effective and sustained scaling pathways.

In the project-based pathway (ASA and CSICAP), implementation is led by project technicians and partner financial institutions that support farmer investment. In the LTAC pathway, delivery is carried out by technical staff from member organizations aligned with scaling strategies co-developed with S4I. In the policy pathway, implementation is channeled through government-linked programs and their associated technical teams.

In terms of relative contribution to scale, the project-based pathway is expected to generate approximately 75% of total outreach, given its access to large-scale investment and established delivery mechanisms. The LTAC pathway contributes an estimated 20%, leveraging its territorial networks and multi-actor coordination platforms. The policy pathway is expected to account for around 5%, primarily through gradual institutional embedding into public programs and long-term scaling processes.

3.2 Theory of Change

If S4I acts as a systemic catalyst—by packaging proven water and soil solutions into context-adapted 'bundles' and integrating them into large-scale investment projects, territorial climate information networks (such as Local Technical Agroclimatic Committees - LTACs), and national policy frameworks—**and if** it simultaneously strengthens the coordination capacity of the agricultural innovation system, **then** current institutional fragmentation and adoption gaps will be overcome; **consequently**, this will enable both key stakeholders and farmers to transition from isolated practices toward the integrated use of climate-smart solutions, achieving a sustained increase in productivity and climate resilience that culminates in the institutionalization of scaling, ensuring that impact endures beyond the lifespan of the projects. **Key Bottlenecks to Scaling and S4I Response**

Despite the availability of validated climate-smart practices, several systemic bottlenecks continue to limit their adoption at scale:

1. Fragmented agricultural innovation systems: Weak coordination among public, private, and community actors limits coherent delivery and scaling.

S4I response: Strengthen coordination mechanisms through articulation with AKIS diagnostics (AW1), strategic partner engagement, and the use of LTACs as territorial platforms for alignment and joint action.

2. Lack of packaged, context-specific solution bundles: Existing practices are often promoted individually rather than as integrated, actionable packages adapted to local conditions.

S4I response: Package validated practices into territorially adapted bundles through participatory processes, ensuring they are actionable, relevant, and scalable across different agroecological and socio-economic contexts.

3. Limited integration into large-scale delivery systems: Proven solutions are not systematically embedded in major investment projects, extension systems, or public programs.

S4I response: Co-design and embed scaling strategies within ongoing large-scale initiatives (e.g., ASA, GCF-CSICAP), as well as within LTAC networks and policy frameworks, to ensure delivery through existing channels.

4. Financial constraints for farmer adoption: Limited access to tailored financial products restricts the uptake of investment-intensive practices such as irrigation and infrastructure.

S4I response: Engage microfinance institutions and cooperatives to co-develop financing mechanisms aligned with solution bundles, enabling farmer-led investment and sustained adoption.

By directly addressing these bottlenecks, S4I ensures that activities are not standalone interventions, but targeted actions that unlock systemic constraints to scaling.

3.3 Key Activities Funded by S4I

The following activities are designed to directly address the key bottlenecks identified above and enable effective scaling across pathways:

- **Pathway strategy design:** Participatory development or strengthen of scaling strategies for project-based, LTAC, and policy pathways, including state-of-the-art analysis of actors, capacities, and bottlenecks.
- **Territorial adaptation and validation:** Co-design and refinement of context-specific bundles with local partners.
- **Partner accompaniment:** Technical guidance to partners across all pathways to operationalize scaling strategies, integrate bundles into training and extension, and strengthen territorial coordination.
- **Evidence generation for scaling:** Demand-driven generation of evidence aligned with the needs of partners across the different scaling pathways, and tailored to their stage of progress in scaling water and soil practices. This may include, for example, the documentation of bundles (e.g., technical manuals), agroecological zoning analyses, and the development of agronomic, economic, environmental, and social evidence to inform and support scaling decisions.
- **Monitoring, evaluation, and adaptive management:** Design (and/or strengthen) and implementation of participatory scaling monitoring systems to track progress, adoption quality, and risks.
- **Knowledge management and Regional Scaling Hub:** Systematization of lessons learned, linkage with the LAC Regional Scaling Hub of S4I, and dissemination of learning products to attract new actors.

4. Partners and Roles

Partner Type	Specific Partners and Quantities	Role in Scaling Pathway
Demand Partners (Articulate farmer needs)	**Government:** MAGA, MARN (Guatemala); MADR, MADS (Colombia). **Local organizations:** ASORECH, CDRO, ADIPAZ. **Producer associations** (linked to GCF-CSICAP project in Colombia).	Define priorities; validate bundles; ensure alignment with national policies and local realities; provide entry points for integration into public programs.
Scaling Partners (Enable and deliver at scale)	**LTAC Network:** 77 committees across 11 countries, engaging over 400 partners. **Project implementers:** CRS (ASA project), GCF-CSICAP project teams. **Finance and private sector:** COOSAJO, FEDECOVERA (Guatemala); Contactar, Popoyán (Colombia). **Extension:** UMATA and EPSAGROS (Colombia).	**LTACs:** Identify and disseminate prioritized bundles. **Project implementers:** Integrate bundles and scaling strategies into large-scale operations. **Finance:** Co-design co-financing mechanisms, expand access to credit. **Extension:** Deliver bundles to last-mile farmers.
Regional and Knowledge Partners	Central American Agricultural Council (CAC), Latin America Climate Action Platform (PLACA), Regional Scaling Hub (LAC).	Provide regional policy alignment; facilitate cross-country learning and replication; channel engagement and knowledge sharing.

Funding Model: Scaling is primarily financed through partner-led investments embedded in large-scale programs such as ASA and CSICAP, which allocate resources for implementation, technical assistance, and farmer outreach. Complementary financing is mobilized through microfinance institutions and cooperatives, enabling farmers to invest in selected components of the bundles. S4I catalytic funding does not finance implementation directly; instead, it supports the critical functions required to enable scaling (such as strategy design, bundle packaging, evidence generation, coordination, and learning) thereby increasing the effectiveness, reach, and sustainability of partner investments.

5. Enabling Environment and Risks

5.1 Opportunities

- **Strong policy alignment:** Bundles directly contribute to national priorities, including Colombia's NDC 3.0, Guatemala's National Irrigation Policy, and the EASAC strategy (in Central America), creating political tailwinds for adoption. **Institutionalized platforms:** LTACs and the Regional Scaling Hub provide ready-made, trusted infrastructure for engagement, delivery, and learning.
- **Existing large-scale projects:** The GCF-CSICAP and CRS-ASA projects offer immediate, funded channels for testing and embedding scaling strategies, ensuring rapid reach.
- **Long-standing partner relationships:** Trust and continuous collaboration with government units, local organizations, and microfinance institutions reduce transaction costs and enable rapid mobilization.

5.2 Challenges and Mitigation

- **Political risk / staff turnover:** Frequent changes in government and decision-makers can disrupt program continuity.

- *Mitigation:* Secure commitments at both national and local levels; anchor practices in existing policies and legal frameworks; build strong relationships with technical staff, whose turnover is lower.
- **Financial sustainability:** Budget cuts and competition for resources may weaken public institutional capacity.
- *Mitigation:* Diversify funding sources (public, international climate finance, microfinance); leverage solid impact evidence to attract new investments; embed cost-recovery mechanisms through cooperatives.
- **Coordination failure / fragmentation:** A recent diagnostic of Guatemala's agricultural innovation system confirmed weak coordination among innovation system actors.
- *Mitigation:* Use the AKIS approach to map actors and incentives, then strategically strengthen networks and broker connections to address fragmentation systematically.
- **Social unrest / operational disruptions:** Strikes or blockades in fragile areas can hinder field activities.
- *Mitigation:* Work through legitimate local organizations; adopt flexible planning and allocate contingency resources.
- **Digital divide:** Limited access restricts timely information in rural areas.
- *Mitigation:* Promote locally adapted, low-tech tools and methodologies alongside digital platforms.

6. Learning and Adaptation

6.1 Monitoring Focus

- **Process outcomes:** Functionality and strengthening of scaling pathways; quality of partner engagement; effectiveness of co-designed scaling strategies.
- **Adoption and behavior change:** Number and profile of farmers adopting bundles; quality of adoption (integrated use vs. isolated practices); farmer perceptions of reduced crop losses, improved food security, and adaptive capacity.
- **System outcomes:** Changes in institutional coordination; integration of bundles into public programs and financing mechanisms; evidence of policy influence.
- **Responsible scaling:** Identifying areas where scaling should not occur; monitoring environmental outcomes (soil moisture, GHG emissions).

6.2 Adaptive Management

Learning will be systematized and fed back into implementation through:

- **Quarterly check-ins** with the S4I AoW2 team to review progress against milestones.
- **Annual reflection workshops** with partners across all pathways to capture lessons, identify bottlenecks, and co-design adaptations.
- **Regular engagement with the Regional Scaling Hub** to share findings and access learning from other tracks.
- **A participatory scaling monitoring system** co-designed with partners will track both quantitative metrics and qualitative insights, enabling real-time course correction.

7. Next Steps and Timeline (2026)

Quarter	Key Activities
Q1	• Finalize detailed work plans for each pathway. • Conduct inception meetings with core partners (CRS, GCF-CSICAP teams, key LTACs). • Initiate AKIS diagnostic in Colombia to map innovation system actors. • Begin state-of-the-art analysis of scaling readiness for each pathway.
Q2	• Co-design scaling strategies for the project-based pathway (ASA, CSICAP) using IPSR where needed. • • Hold first pathway-specific coordination meetings with finance partners (COOSAJO, Contactar). • Develop and disseminate first communication products on the scaling track's launch.
Q3	• Begin accompaniment and technical guidance to partners in implementing scaling strategies. • Initiate territorial adaptation and validation of bundles with local organizations. • Document early lessons and evidence from project-based scaling. • Present progress at the Regional Scaling Hub knowledge exchange event.
Q4	• Achieve 2026 target of 12,000 farmers reached. • Formalize engagement agreements with at least 10 new strategic partners. • Complete first annual MEL report documenting adoption, lessons, and adaptive adjustments. • Produce and disseminate learning products (technical notes, policy briefs) targeting ministries and financiers.

Communications and Knowledge: Key products will include scaling pathway strategy briefs, technical manuals for bundles, policy briefs for national and regional bodies, and regular updates shared via the Regional Scaling Hub and CGIAR channels. Communications support will be strategically deployed to coincide with major milestones (e.g., reaching 12,000 farmers, policy engagement wins) to amplify success stories and attract new partners.

8. Contact

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